Experiment 6

AIM : Setup Nginx to handle proxy requests and load balancing.

```
C:\Users\DELL>mkdir load-balance-demo
C:\Users\DELL>cd load-balance-demo
C:\Users\DELL\load-balance-demo>npm init -y
Wrote to C:\Users\DELL\load-balance-demo\package.json:
{
  "name": "load-balance-demo",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
C:\Users\DELL\load-balance-demo>npm install express
added 65 packages, and audited 66 packages in 5s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
```

Make Server 1 Folder

```
C:\Users\DELL\load-balance-demo>mkdir server1
C:\Users\DELL\load-balance-demo>cd server1
C:\Users\DELL\load-balance-demo\server1>npm init -y
Wrote to C:\Users\DELL\load-balance-demo\server1\package.json:
{
  "name": "server1",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
C:\Users\DELL\load-balance-demo\server1>npm install express
added 65 packages, and audited 66 packages in 2s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
```

```
C:\Users\DELL\load-balance-demo\server1>notepad server.js
C:\Users\DELL\load-balance-demo\server1>node server.js
Server running on port 3000
```

Response from Server 1 - Port 3000

Make Server2 Folder

```
PS C:\Users\DELL> cd load-balance-demo
PS C:\Users\DELL\load-balance-demo> mkdir server2
PS C:\Users\DELL\load-balance-demo> cd server2
PS C:\Users\DELL\load-balance-demo\server2> npm init -y
Wrote to C:\Users\DELL\load-balance-demo\server2\package.json:
{
  "name": "server2",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}
PS C:\Users\DELL\load-balance-demo\server2> npm install express
added 65 packages, and audited 66 packages in 6s
22 packages are looking for funding
  run `npm fund` for details
found 0 vulnerabilities
PS C:\Users\DELL\load-balance-demo\server2> notepad server.js
PS C:\Users\DELL\load-balance-demo\server2> node server.js
Server running on port 3001
```

Response from Server 2 - Port 3001

Now Run Nginx

```
C:\nginx\nginx-1.28.2>nginx -t
nginx: the configuration file C:\nginx\nginx-1.28.2/conf/nginx.conf syntax is ok
nginx: configuration file C:\nginx\nginx-1.28.2/conf/nginx.conf test is successful

C:\nginx\nginx-1.28.2>nginx
```

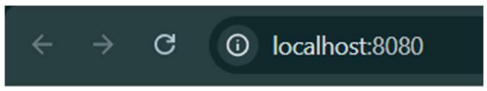
Response from Server 1 - Port 3000

Response from Server 2 - Port 3001

Modify refresh time in Server 1.

```
C:\nginx\nginx-1.28.2>nginx -s stop
C:\nginx\nginx-1.28.2>nginx -t
nginx: the configuration file C:\nginx\nginx-1.28.2/conf/nginx.conf syntax is ok
nginx: configuration file C:\nginx\nginx-1.28.2/conf/nginx.conf test is successful

C:\nginx\nginx-1.28.2>nginx
```



Slow Server 1- Port 3000

NGINX still keeps sending every alternate request to it.

Modify nginx.conf file:

```
C:\nginx\nginx-1.28.2>nginx -s stop
C:\nginx\nginx-1.28.2>nginx -t
nginx: the configuration file C:\nginx\nginx-1.28.2/conf/nginx.conf syntax is ok
nginx: configuration file C:\nginx\nginx-1.28.2/conf/nginx.conf test is successful

C:\nginx\nginx-1.28.2>nginx
```

Response from Server 2 - Port 3001

First show Fast Server.